

Research

Enhancing energy security and cost efficiency in Nigerian higher institutions via hybrid renewable energy systems

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Abstract

The growing global energy demand, fossil fuel depletion, and environmental concerns highlight the need for sustainable energy alternatives. In Nigeria, persistent power shortages critically affect key sectors, including education. This study conducts a techno-economic analysis of a hybrid photovoltaic-diesel-battery system to enhance energy reliability and sustainability in underserved academic institutions. Using a Nigerian College of Education as a case study, the system was designed and optimised using Hybrid Optimisation of Multiple Energy Resources (HOMER) software, incorporating real institutional load data and local economic inputs. The proposed system comprises a 1000 kVA diesel generator, a 32 kW solar PV array (sized under spatial and economic constraints), a converter, and battery storage with a usable capacity of approximately 43,758 kWh. Two dispatch strategies—Cycle Charging (CC) and Load Following (LF)—were evaluated. CC outperformed LF, achieving a net present cost (NPC) of \$57.1 million and a levelized cost of energy (COE) of \$0.241/kWh, compared to the diesel-only base case with an NPC of \$99.5 million and COE of \$0.419/kWh. The hybrid configuration achieved a 55% renewable fraction, reducing fuel consumption and avoiding roughly 2034 tonnes of CO₂ emissions annually. Sensitivity analysis, conducted across diesel prices (\$1.0–\$1.8/L) and capital cost multipliers (0.25–1.0), showed NPCs ranging from \$4.42 M to \$6.26 M and COEs from \$0.187 to \$0.264/kWh, highlighting the impact of cost variables on system feasibility. The analysis assumes a diesel price of \$1.6/L, a 12.5% discount rate, and a 25-year project life. The findings demonstrate that integrating solar PV, diesel generation, and battery storage is a viable strategy for improving energy access, reducing operational costs, and lowering emissions in educational institutions. The methodology is adaptable to similar off-grid or unreliable-grid settings in sub-Saharan Africa, Southeast Asia, and other energy-constrained regions, providing a useful reference for institutional energy planning and renewable energy policy development. *Clinical Trial Number*: Not applicable.

Keywords Hybrid energy system · Photovoltaic-diesel-battery · Renewable energy optimization · Sustainable power supply · Techno-economic analysis

1 Introduction

Access to reliable and affordable electricity remains a pressing global concern, particularly in developing countries with underdeveloped or poorly maintained power infrastructure. Despite significant technological advances and global commitments toward universal energy access, over one billion people still lack access to dependable electricity services [1]. This energy gap severely affects social and economic development, with academic institutions among

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the hardest hit. The availability of stable electricity is fundamental for running laboratories, ICT infrastructure, digital learning platforms, administrative operations, and research activities [2]. Inadequate power supply undermines the quality of education, restricts academic innovation, and stifles institutional growth.

In Nigeria, the energy crisis remains a major impediment to national development. Despite having the largest economy in Africa, the country suffers from chronic electricity shortages due to insufficient generation capacity, ageing infrastructure, and frequent grid collapses. Unreliable grid power primarily affects public universities, polytechnics, and especially colleges of education. These institutions frequently resort to diesel generators to maintain operations, but such reliance is unsustainable due to the escalating cost of diesel, environmental degradation, noise pollution, and high maintenance burdens [3] (Mas'ud et al., 2024). As the global call for sustainable and clean energy transitions intensifies, integrating renewable energy technologies into Nigeria's energy portfolio is both an environmental imperative and a strategic necessity.

Renewable energy sources (RES)—such as solar, biomass, wind, and hydro—offer a viable path toward achieving energy sustainability. Among these, solar photovoltaic (PV) systems have gained significant attention due to their scalability, environmental friendliness, and rapidly decreasing installation costs [4]. However, solar PV systems are subject to weather-induced intermittency and daily solar radiation variability, which limits their reliability in standalone applications [5]. To overcome this challenge, hybrid energy systems (HES) have been developed to combine multiple energy sources—typically solar PV, diesel generators, and battery storage—into an integrated solution that balances availability, cost, and environmental impact (Abirami et al., 2025) [6].

Numerous studies have explored the application of hybrid renewable energy systems (HRES) in different contexts. Baseer et al. [1] examined PV-diesel-wind systems for residential electrification in Saudi Arabia, demonstrating technical feasibility and long-term economic benefits. Murugaperumal et al. [7] designed an integrated biomass-wind-solar system for rural India, emphasising system reliability and grid independence. Akram et al. [8] incorporated demand-side management into HRES optimisation, highlighting operational cost savings and demand flexibility. Liu et al. [9] compared PV-battery and PV-wind-battery systems for high-rise buildings and found that hybrid configurations significantly reduce reliance on conventional grid power while improving energy efficiency.

In the African context, hybrid systems have proven effective in various sectors. Olusayo and Gideon [3] evaluated solar-diesel-battery microgrids for Nigerian universities, noting the sensitivity of system performance to diesel price volatility. Mohammed et al. (2025) and Benzzine et al. [10] modelled microgrids for off-grid communities using tools such as TRNSYS and HOMER, affirming the cost-effectiveness of hybrid configurations. Gaurav et al. [11] and Woldegiyorgis et al. [12] applied HOMER to agricultural and healthcare facilities, showcasing trade-offs between capital investment and renewable energy penetration. Tural and Akar [13] used RETScreen and MATLAB for load and economic modelling but highlighted the absence of dispatch control and sensitivity tools.

Despite these advances, there remains a significant research gap in the application of hybrid systems to Colleges of Education—institutions that are equally vulnerable to energy unreliability yet have received minimal attention in both national electrification efforts and academic research. While some studies have analysed university campuses [3, 6], very few have addressed other tiers of tertiary education. The Energising Education Programme (EEP), spearheaded by Nigeria's Rural Electrification Agency (REA), has focused on select federal universities and teaching hospitals (EEP, 2021), leaving colleges of education excluded from renewable energy transitions.

Moreover, many existing studies rely on standardised or assumed load profiles, generic cost data, and idealised system configurations. For instance, Babaei et al. [14] and Muskan and Kaur [15] simulated hybrid systems for rural clinics but did not incorporate measured consumption data. Similarly, dispatch strategies like Cycle Charging (CC) and Load Following (LF) are often inadequately compared, despite their significant implications for energy cost and battery lifespan. Sensitivity analyses—critical for understanding the impact of fluctuating diesel prices and component costs—are frequently omitted or narrowly conducted. These limitations reduce the practical applicability of findings for real institutional planning.

To fill these gaps, this study proposes a detailed simulation and optimisation of a hybrid PV-diesel-battery system tailored to the College of Education, Akwanga, Nasarawa State, Nigeria. The institution's current energy model—based solely on diesel generators due to unreliable grid supply—is analysed as the base case. Real institutional load data were collected to accurately represent energy consumption patterns. The hybrid system design was conducted using the HOMER Pro software, selected for its robust capabilities in simulating off-grid and grid-tied systems, performing techno-economic optimisation, and enabling sensitivity testing across multiple variables [2, 16, 17].

The study models several configurations of PV, diesel, and battery components to identify the most cost-effective and environmentally sustainable system. It evaluates two dispatch strategies—cycle charging and load following—to

determine which yields optimal performance in terms of net present cost (NPC), levelized cost of energy (COE), and fuel consumption. A sensitivity analysis is conducted to assess system resilience under varying diesel prices and capital costs, while environmental impact is quantified through estimated greenhouse gas (GHG) emission reductions. These methodological enhancements enable a more holistic understanding of hybrid energy systems in education-based microgrids.

The novelty of this work lies in its exclusive focus on a college of education—a segment that has been largely neglected in energy policy and scholarly research. Unlike prior studies, this research uses field-based load measurements, localised cost assumptions, dispatch strategy comparison, and economic-environmental sensitivity analyses. It also addresses practical constraints such as limited rooftop area for PV installation and daily variation in electricity usage across academic and non-academic functions. The findings offer a transferable framework that can be replicated by other educational institutions in Nigeria and across sub-Saharan Africa.

In conclusion, this study makes a significant contribution to the literature on institutional energy planning in developing countries. It demonstrates that hybrid PV-diesel-battery systems can substantially enhance energy reliability, reduce operational costs, and lower environmental impact in Nigerian Colleges of Education. The results have direct implications for policymakers, institutional managers, energy investors, and development agencies aiming to bridge the energy access gap in education. The integration of real-world data, advanced modelling, and context-specific analysis positions this research as a critical resource for scaling sustainable energy transitions in underserved academic sectors.

The overarching objective is to design, simulate, and optimise a hybrid energy system for the College of Education, Akwanga, using HOMER Pro. Specific aims include determining the institution's energy demand profile based on real-time data, evaluating multiple hybrid configurations, comparing dispatch strategies, and conducting a comprehensive techno-economic and environmental analysis. This integrated approach not only responds to local energy challenges but also provides actionable insights for wider adoption of hybrid renewable energy solutions in educational facilities across developing regions.

2 Methods

2.1 Case study description

The HES under consideration is designed for the College of Education, Akwanga, Nasarawa State, located at latitude 8.9°N and longitude 8.4°E. This autonomous educational institution operates from 8:00 AM to 4:00 PM. The peak power demand reaches 3804.12 kW, while the average daily energy consumption is 15,693 kWh/day. Due to the unreliability of the existing 11 kV high-tension (HT) supply from Lafia, the proposed HES aims to provide a more sustainable and dependable off-grid energy alternative. The HOMER simulation excluded grid interaction due to persistent instability in the local 11 kV high-tension supply.

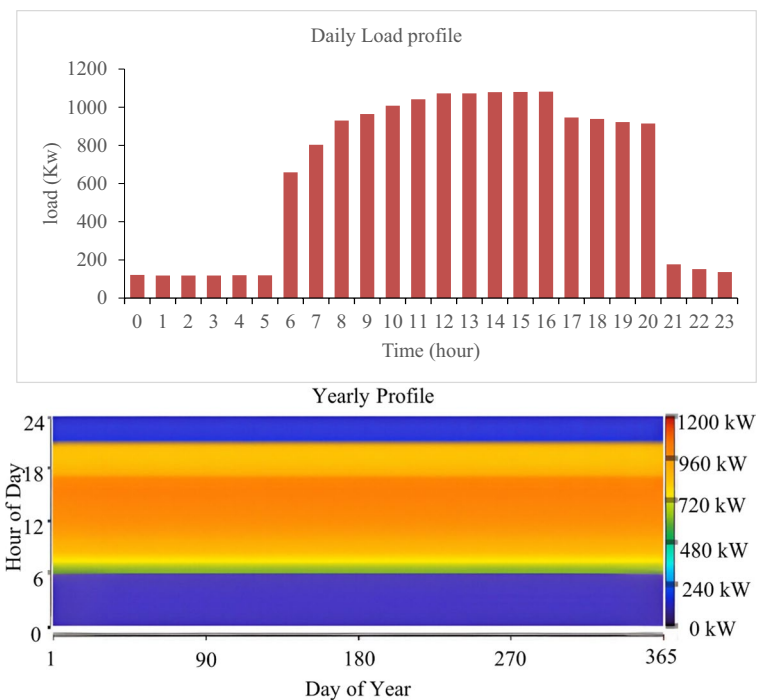
2.2 Load profile

The institution's energy consumption pattern is distinct from residential, commercial, and industrial usage, varying according to semester schedules, vacations, weekdays, and weekends. Energy demand in research laboratories and for lighting fluctuates throughout the year. Data recorded using an Owl meter indicate an average daily energy demand of 15,693 kWh, which serves as the synthetic load input for HOMER software analysis. The peak demand is 1081.6 kW, with a load factor of 0.6, day-to-day random variability of 20%, and a time-step variability of 10%. The average hourly, daily, and seasonal load profiles are illustrated in Fig. 1.

2.3 Determination of solar radiation

The output power of RES is significantly influenced by both system component design and environmental factors. In the case of solar photovoltaic systems, solar radiation and ambient temperature are key factors influencing performance. The planned HES relies on Global Horizontal Irradiation (GHI) data, incorporating the yearly variations in monthly average solar radiation, clearness index, and ambient temperature.

Fig. 1 Load Profile of College of Education Akwanga



For this study, solar radiation data for the chosen location—College of Education, Akwanga, Nasarawa State (latitude 8.9°N, longitude 8.4°E)—were sourced from the National Renewable Energy Laboratory (NREL), a comprehensive repository for RES-related datasets. The highest recorded solar insolation at this location occurs in March, reaching 6.15 kWh/m²/day, while the lowest is observed in July at 4.48 kWh/m²/day. Figure 2 presents the variations in solar radiation, clearness index, and ambient temperature as retrieved from the NREL database for the College of Education, Akwanga, Nasarawa State.

2.4 Description of hybrid energy system components

The proposed HES comprises a solar photovoltaic (PV) system, a diesel generator, a converter, battery storage, and an AC electrical load. A techno-economic analysis is conducted using various performance and evaluation metrics to assess its feasibility. HOMER software is used to enhance the system's economic and technical performance, incorporating component specifications and costs obtained through a market survey. The proposed HES for the College of Education, Akwanga, Nasarawa State, includes a 1000 kVA diesel generator and a 32 kW solar PV system, along with batteries and a converter, depicted schematically in Fig. 3.

The battery storage system consists of ABB PS-BatME2 lithium-based batteries, with a total installed capacity of 43,758 kWh. These batteries were selected for their deep-cycle capability, high efficiency, and compatibility with the selected converter system. The HOMER simulation assumed a round-trip battery efficiency of 90%, a depth of discharge (DOD) of 80%, and a unit capital cost of \$150/kWh, based on current market surveys.

Although the peak load of the institution is 1081.6 kW, the HOMER-optimised PV array size was limited to 32 kW. This relatively small size results from several practical constraints: (i) the high capital cost of PV modules and associated components, (ii) limited rooftop and land area available for installation at the time of system design, and (iii) HOMER's economic optimisation that prioritised systems with the lowest NPC and COE under budgetary constraints. The PV contribution was thus economically rather than technically maximised.

The converter selected for the hybrid system is the ABB PSTORE-PCS, with a rated capacity of 1065.35 kW. This power conversion system enables bidirectional power transfer between the DC side (PV and battery) and the AC load, ensuring system reliability and synchronisation. The HOMER software assumed a conversion efficiency of 95%, a capital cost of \$300/kW, and a lifespan of 15 years. The converter also supports grid-forming operation during islanding or off-grid conditions.

Fig. 2 Solar resources in College of Education Akwanga (a) solar radiation and clearness index and (b) temperature of the chosen location

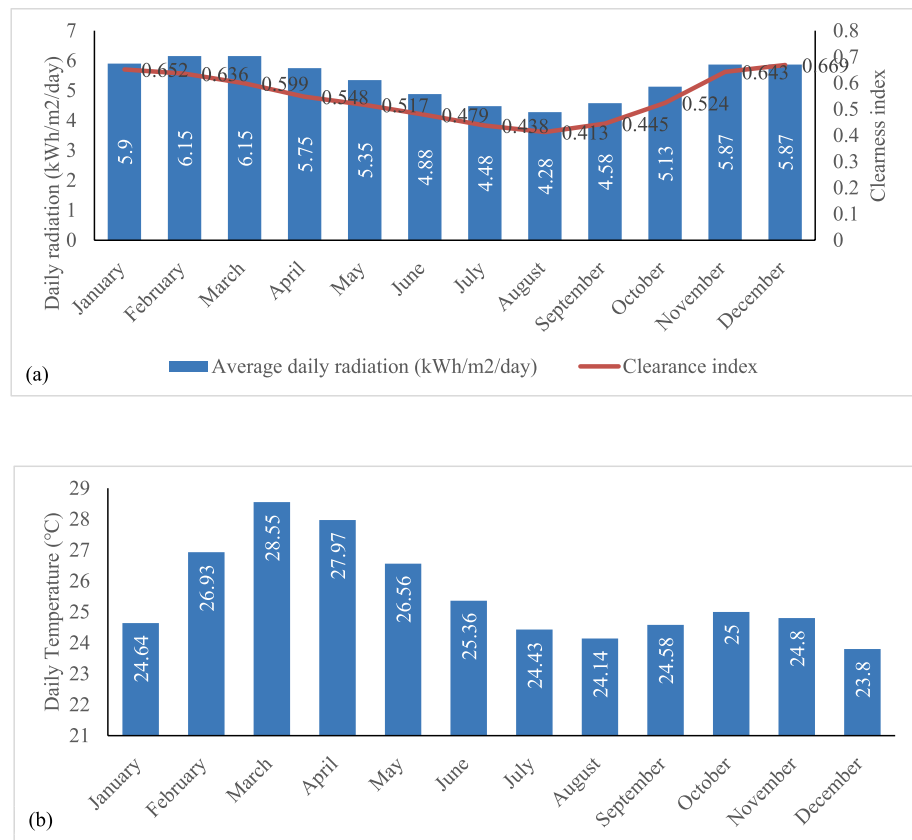
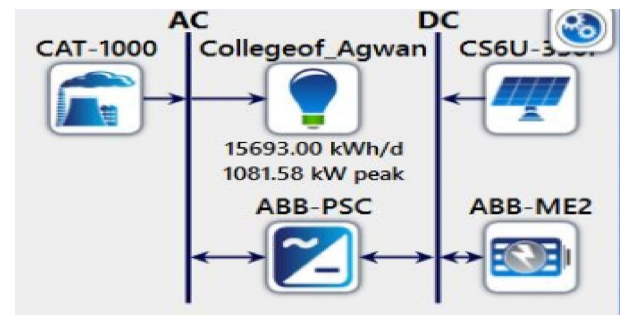


Fig. 3 Schematic Layout of the Proposed System



2.5 Determination of photovoltaic array size

The solar PV module serves as the primary and initial component of the proposed HRES. It is composed of PV cell circuits enclosed within an environmentally friendly laminate. These solar panels capture solar radiation and convert it into electrical energy. The efficiency of the PV system is influenced by factors such as the amount of incident global solar radiation, the temperature of the PV panels, and the PV conversion factor [18]. Accordingly, Eq. (1) was employed to calculate the hourly power output of the PV system [19].

$$P_{pv} = Y_{pv} f_{pv} \left[\frac{G_T}{G_{T,STC}} \right] [1 + \alpha_p (T_c - T_{c,STC})] \quad (1)$$

where Y_{pv} represents the rated capacity of the PV array under standard test conditions (kW), G_T denotes the solar radiation incident on the PV array at the given time step (kW/m²), $G_{T,STC}$ refers to the incident radiation under standard test conditions (1 kW/m²). Additionally, α_p is the power temperature coefficient (%/°C), T_c indicates the PV cell temperature at the current time step (°C), while $T_{c,STC}$ corresponds to the PV cell temperature under standard conditions (25 °C).

2.6 Battery bank sizing

Solar PV systems are best suited for deep-cycle batteries, as they can be discharged to low energy levels and rapidly recharged over extended periods. The battery capacity should be sufficient to sustain operations during night-time, cloudy, rainy, and dusty days.

For economic analysis in HOMER, the following assumptions were made for the battery bank: capital cost = \$150/kWh, replacement cost = \$100/kWh, O&M cost = \$10/kWh/year, lifespan = 10 years. These parameters were used to calculate the total lifecycle cost and replacement cycles of the battery storage system. The required battery capacity is determined using Eq. (2) [20]:

$$E_{\text{BAT}} = \frac{E_L \times d}{\eta_{\text{conv}} \times V_s \times \eta_{\text{BAT}} \times \text{DOD}} \quad (2)$$

where E_L is the required daily load (kWh), d denotes the backup duration (days), η_{conv} is the converter efficiency, V_s is the system voltage (V), η_{BAT} is the battery efficiency, and DOD is the depth of discharge (%).

The quantity of batteries, N_{BAT} is determined by [13, 21, 22]:

$$N_{\text{BAT}} = \frac{\text{battery bank capacity (kWh)}}{\text{battery capacity (Ah)}} \quad (3)$$

A power electronic converter with a high operational efficiency of 95% was employed in this study, serving dual functions as both a rectifier (AC to DC conversion) and an inverter (DC to AC conversion). This bidirectional capability ensures seamless integration and energy flow management within the hybrid power system.

2.7 Diesel generator

Diesel generators are commonly combined with renewable energy sources to enhance system reliability. The diesel generator's fuel consumption can be approximated using the Eq. (4) (Camargo et al., 2019):

$$F_c = C_g P_o + K_g P_r \quad (4)$$

where P_o represents the generator's rated power, P_r is the operating power output, K_g is the fuel curve slope (litres/kWh), and C_g is the intercept coefficient of the fuel curve (litres/kWh).

2.8 Techno-economic optimization with hybrid optimization of multiple energy resources software

The performance of a hybrid power system relies on precise technical design and economic evaluation. Proper sizing of system components ensures efficient operation year-round. HOMER software is commonly used for modelling and optimising hybrid power systems. The key goal is to reduce electricity costs by minimising demand charges while maximising the use of renewable energy sources, ultimately decreasing greenhouse gas emissions.

HOMER evaluates economic feasibility through essential financial performance metrics, COE, NPC, return on investment (ROI), payback period (PBP), and internal rate of return (IRR) [23]. In this research, HOMER Grid was utilised to model the proposed hybrid power system, maintaining an hourly energy balance between supply and demand throughout the year (8760 h).

Compared to other simulation tools, such as RETScreen and TRNSYS, HOMER provides a more integrated platform for hybrid energy system design by combining technical simulation, dispatch strategy modelling, and detailed economic optimisation. While RETScreen offers feasibility analysis, it lacks time-series dispatch simulation. TRNSYS offers advanced thermal and dynamic modelling but requires extensive technical setup and programming knowledge. MATLAB/Simulink allows detailed custom models but does not natively support hybrid optimisation without extensive scripting and lacks built-in financial modules.

HOMER's built-in sensitivity analysis, ease of use, and support for hybrid configurations make it ideal for rural electrification, microgrid design, and institutional energy planning in developing countries. Its effectiveness in handling hourly dispatch, cost optimisation, and CO₂ reduction under varying fuel and component price scenarios makes it a robust tool for this study.

In HOMER, two dispatch strategies were analysed: LF and CC. Under the LF strategy, the generator supplies just enough power to meet the load, with excess demand met by stored energy. In contrast, CC runs the generator at its rated capacity whenever it is on, directing surplus power to charge the batteries. CC is generally more efficient in terms of generator fuel usage and battery management when load profiles are variable and storage capacity is available. For systems incorporating energy storage devices such as batteries or fuel cells, or those utilising diesel generators, HOMER determines whether to charge or discharge storage and whether to operate the generator in each time step. The software utilises two optimisation algorithms: a preliminary search algorithm that assesses all possible system configurations and a proprietary derivative-free method to determine the most cost-efficient arrangement. HOMER prioritises system designs based on the lowest NPC, enabling comparisons between alternatives [24].

Furthermore, HOMER considers uncertainties like interest rates, solar radiation, fuel costs, and wind speed. Sensitivity analysis is performed by inputting multiple values for each parameter, enabling comprehensive optimisation for varying conditions. Converter performance and sizing were optimised to ensure system stability and efficient power management. The size was selected based on the peak AC demand and PV/battery contribution limits to the AC bus. HOMER Pro conducts techno-economic optimisation using a derivative-free, brute-force simulation algorithm. The primary objective function is to minimise the Net Present Cost (NPC) of the system, expressed as [25]:

$$NPC = \frac{C_{ann,tot}}{CFR(i, N)} \quad (5)$$

where $c_{ann,tot}$ denotes the total annualized cost ($\frac{\$}{yr}$), $CFR(i, N)$ represent the capital recovery factor. The capital recovery factor (CRF) is used to calculate the present value of the annual revenue and expenditure stream, as shown in Eq. (6) [25]

$$CFR(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (6)$$

where: i denotes Real interest rate (%), N is Project life (year). The unit cost of energy is defined as the average cost per kWh of usable energy produced by the electricity generation system, as calculated by HOMER. It is explained in Eq. (7) [25].

$$COE = \frac{C_{ann,tot}}{E_{prim}} \quad (7)$$

where: COE represents cost of energy (\$/kWh), and E_{prim} represents the primary load being served. (kWh/year) [25, 26]

The optimisation process involves evaluating thousands of feasible system configurations by varying decision variables such as PV size, battery capacity, diesel generator size, and dispatch strategy. The process is subject to constraints including: minimum renewable energy fraction, maximum allowable capacity shortage, and component lifetimes and operational limits.

HOMER ranks each configuration based on NPC, using sensitivity analysis to evaluate robustness under variable input conditions (fuel price, component cost, etc.). This multi-objective, constraint-based simulation allows for selecting the most cost-effective and technically viable system.

Table 1 Economic metrics of the optimised system as compared to diesel generator alone

Economic metrics	Base case	Lowest cost system (\$)
NPC	99.5M (\$)	57.1 M (\$)
Initial capital	773.28 (\$)	2.44 M (\$)
O&M	2.38 M/yr (\$)	1.32 /yr (\$)
COE	0.419/ kWh	0.241 /Kwh
IRR	–	65%
ROI	–	60%
Simple payback	–	1.53 yr

Table 2 Cost summary by components based on net present cost

CS6U-330P (kW)	CAT-1000 (kW)	ABB-ME2	ABB-PSC (kW)	Dispatch architecture	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)	Ren Frac (%)	CO ₂ (kg/yr)
2000	800	43,758	1065.348	CC	57.1 M	0.240779	1,319,694	2,442,491	50.69	1,948,927
	800	13,068	649.1062	CC	99.5 M	0.419333	2,382,906	773,286.1	0	3,983,094

3 Results and discussion

3.1 Techno-economic cost parameters of the optimal system

The key cost parameters used to evaluate the HES are the NPC and the COE. These parameters assist in selecting the most cost-effective system configuration while considering the constraints: include an annual capacity shortage limit of 10%, a minimum renewable energy contribution of 40%, a diesel price of \$1.6 per litre, and a primary load of 15,693 kWh/m²/day. Additionally, economic constraints, including a nominal discount rate of 12.5% and a system lifetime of 25 years, were considered. A method for decision-making based on multiple criteria was employed to evaluate different system configurations and determine the most suitable design for the site. The economic parameters of the hybrid system are presented in Table 1.

The base case, representing a diesel-only system currently in use at the institution, has an NPC of \$99.5 million, with a COE of \$0.419/kWh. In contrast, the optimal HES system has an NPC of \$57.1 million and a COE of \$0.241/kWh when operated using the CC dispatch strategy, which yielded a more efficient system configuration compared to the LF strategy, which was not among the feasible results.

Operational and maintenance (O&M) costs as well as the initial capital cost for the base case system are \$2.38 and \$773,280 million per year, respectively. For the optimal system, these costs are \$2.44 million and \$1.32 million per year under the CC strategy. Additionally, the optimal system exhibits an IRR of 65%, an ROI of 60%, and a simple payback period of 1.53 years, as presented in Table 1.

The optimised hybrid system demonstrates significant financial savings compared to the diesel-only base case. While the base case has a lower initial capital cost of \$773,286, it incurs a high annual operating cost of \$2.38 million due to diesel fuel consumption. In contrast, the hybrid system's higher capital investment of \$2.44 million is offset by lower annual operating costs of \$1.32 million, resulting in a lifecycle cost savings of over \$42 million across 25 years.

The system's Return on Investment (ROI) of 60%, Internal Rate of Return (IRR) of 65%, and a simple payback period of 1.53 years indicate excellent economic performance. These values suggest that the upfront investment can be recovered within 2 years, after which the system continues to generate cost savings, making it a financially sound option for educational institutions with constrained operating budgets.

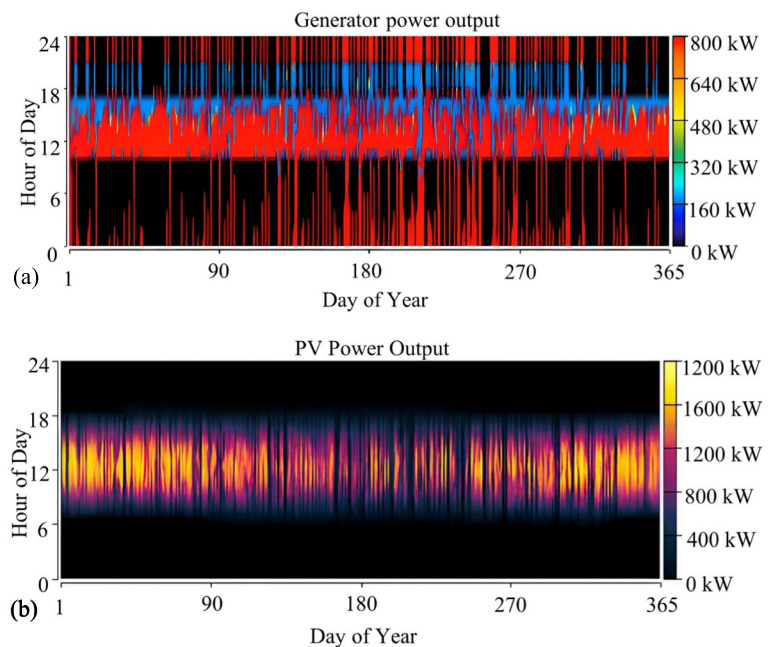
Additionally, when compared with similar systems in developing country contexts (e.g., [27, 28]), the COE of \$0.241/kWh is competitive, especially considering current diesel price fluctuations in Nigeria.

Table 2 details the cost parameters of individual system components based on their NPC values. The NPC accounts for capital investment, salvage value, fuel expenses, O&M costs, and replacement costs, all computed using HOMER Pro software. The diesel generator has the highest capital cost among the HES components, with elevated O&M costs due to fuel expenses. Battery replacement costs amount to \$2,383,730.21, while the PV system incurs no replacement costs since its lifespan matches the project's duration. The total project cost is \$57,110,693.70, with the diesel generator accounting for the highest cost at \$48,403,921.60, while the PV system has the lowest cost at \$350,000. Given that NPC is influenced by multiple external factors beyond equipment purchase rates, the final selection of the optimal system architecture was based on both NPC and capital cost comparisons.

Table 3 Optimised hybrid system for the simulation of photovoltaic (Canadian Solar Max Power CS6U-330P), Generator (CAT-1000kVA-50Hz-PP) using Diesel, Storage (ABB PS-BatME2), and Converter (ABB PSTORE-PCS)

Component	Capital (\$)	Replacement (\$)	Operation and maintenance (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
ABB PS Bat ME2	1,702,886.33	2,383,730.21	4,296,032.21	0.00	(842,338.64)	7,540,310.1
ABB PSTORE-PCS	319,604.51	549,783.10	441,319.75	0.00	(494,245.34)	816,462.0
Canadian Solar Max Power CS6U	350,000.00	0.00	0.00	0.00	0.00	350,000.0
CAT-1000kVA-50Hz-PP	70,000.00	134,851.71	5,553,839.54	42,767,236.64	(122,006.29)	48,403,921.6
System	2,442,490.84	3,068,365.05	10,291,191.50	42,767,236.64	1,458,590.26	57,110,693.7

Fig. 4 (a) Optimal System's Generator Power Output (b) Optimal System's PV Power Output



3.2 Detailed description of CS6U-330P/CAT1000/ABB-ME2/ABB-PSC system

Table 3 summarises the configuration of the optimal system, which consists of CS6U-330P (2000 kW), CAT1000 (800 kW), ABB-ME2 (43,758), and ABB-PSC (1065.35) components, optimised using HOMER's Cycle Charging strategy. A total of 31,426 system configurations were simulated, with infeasible options discarded. The final selection includes two optimised cases: the base case and the most economically efficient system.

NPC and COE were the primary criteria for comparison, guiding consumer selection of the best system architecture. The optimal system has an NPC of \$57.1 million, while the highest NPC, at \$99.5 million, corresponds to the CAT1000 (800 kW)/ABB-ME2 (13,068)/ABB-PSC (649.11 kW) configuration.

As shown in Table 1, the base case has a lower initial capital cost but significantly higher O&M and COE values compared to the CS6U-330P-based optimal system. The lowest-cost configuration has an ROI of 60% and a simple payback period of 1.53 years when operated under the CC dispatch strategy. Jagadeesh et al. [27] reported similar findings, where an optimised system exhibited a high initial capital cost but significantly lower O&M and COE, aligning with the present study.

Table 3 presents the breakdown of capital, replacement, and O&M costs over the 25-year project period. The PV array accounts for 25.4% of the capital cost, followed by batteries (18.7%), the converter (7.1%), and the diesel generator (12.6%). However, the generator dominates O&M costs due to frequent refuelling and servicing requirements. Battery replacement was scheduled twice, based on a 10-year lifespan assumption.

Table 3 further illustrates that the diesel generator, while necessary for backup, contributes the largest share of lifecycle cost due to recurring fuel and maintenance expenses. Conversely, the PV and battery systems, despite their upfront costs, offer long-term stability and a lower operational burden. This reinforces the value of investing in renewable components to reduce dependence on volatile fuel markets.

These costs were used in HOMER to calculate the Net Present Cost and Levelized Cost of Energy. The system's total life cycle cost reflects a balance between higher upfront capital and significantly reduced fuel and O&M expenditures.

3.3 Component contributions in the hybrid renewable energy system

The annual power outputs of the PV (CS6U-330P) and generator (CAT1000) are presented in Fig. 4. The hybrid system integrates PV, DG, batteries, and converters using the HOMER cycle charging strategy. Figure 4a details generator performance, showing 4469 operational hours per year, 310 starts annually, an operational life of 20.1 years, a capacity factor of 40.3%, and a specific fuel usage of 0.261 L/kWh.

Fig. 5 **a** Monthly Electric Production of the Optimal System **b** Excess and Unmet Electrical Production of the HRES System

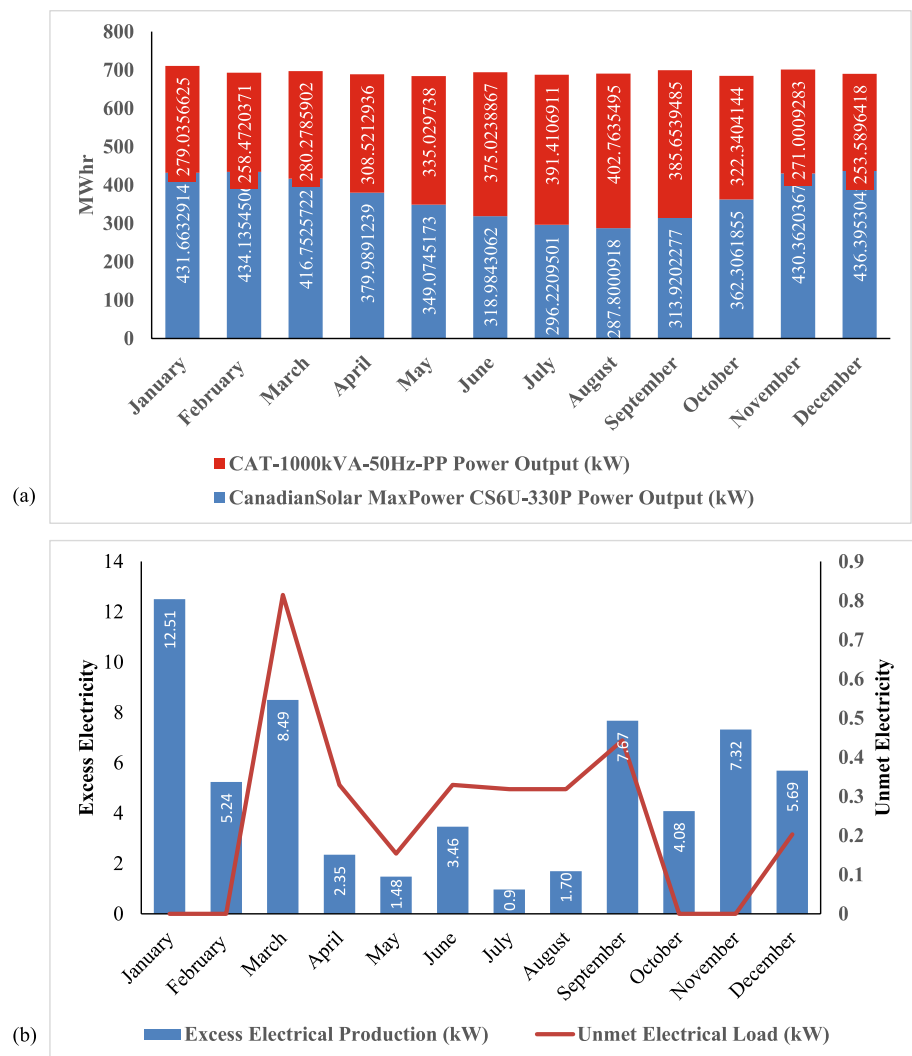


Figure 4b illustrates the annual PV output, with a rated capacity of 2000 kW and an annual generation of 3,250,145 kWh. The mean daily output is 8905 kWh, with an operational duration of 4471 h per year and a capacity factor of 18.6%. These values align with findings by Salameh et al. [28], who reported a capacity factor of 18.2% for a similar system. PV generation follows a diurnal cycle, peaking between 12:00 and 13:00 daily, with seasonal variations influenced by solar insolation levels.

3.4 Power consumption of the hybrid renewable energy system

The HRES serves the college's AC primary load, integrating diesel generators (DGs) with RES. Load power consumption fluctuates between on-grid and off-grid operations based on available energy sources. System optimisation was conducted by evaluating cost parameters and the electrical power output of each generation source.

Renewable energy generation is prioritised due to its lower emissions and sustainability benefits. Figure 4a illustrates the power supply from renewable sources, highlighting that solar energy is the dominant contributor. The HRES comprises a diesel generator and PV system, with total average monthly energy outputs remaining relatively stable throughout the year. However, PV generation declines slightly during the rainy season due to reduced solar irradiance.

Fig. 6 **a** Yearly Diesel Generator Fuel Consumption **b** Greenhouse Gases (GHG) Emissions of the HRES

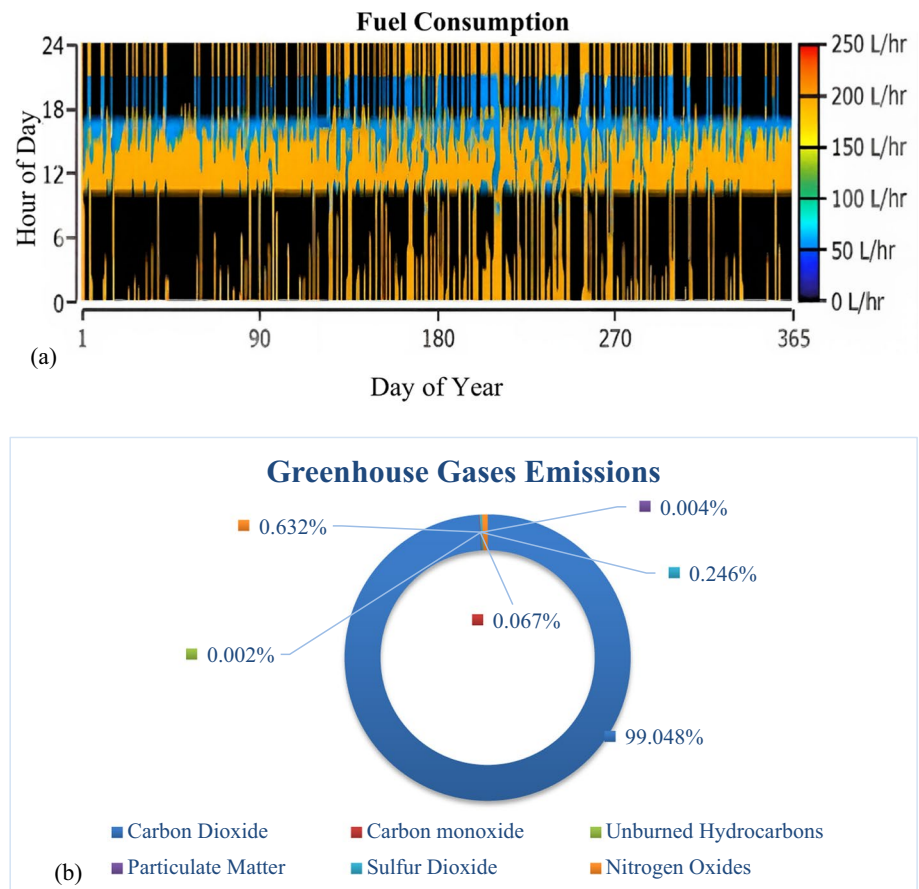


Figure 5a shows that PV generation peaks in June, July, and August, while diesel generator output fluctuates, decreasing in November, December, February, and March, and peaking in July and August. Total annual electricity generation is 6,073,251 kWh, with PV contributing 55% and the diesel generator 45%.

Excess and unmet power generation are presented in Fig. 5b. Surplus energy, derived from both PV and DG, can be utilised for non-essential or deferrable loads. Although minor unmet loads exist, they have a negligible impact on primary load requirements. Battery storage facilitates energy management by storing excess power for AC/DC loads or serving as an independent power source. The off-grid HES functions as a self-sufficient system, where the cycle charging dispatch strategy ensures efficient DG operation and battery storage utilisation.

The renewable fraction (RF) of the optimised system is 55%, corresponding to a PV generation of 3,250,145 kWh/year, while the diesel generator contributes 2,823,106 kWh/year with an RF of 45%. Total system utilisation for AC loads is 5,725,807 kWh/year.

3.5 Fuel consumption and emissions

Figure 6 depicts the annual fuel consumption of the CAT-1000 kVA diesel generator, with a total fuel usage of 737,431 L. Average hourly fuel consumption is 84.2 L, translating to a daily consumption of 2020 L.

Compared to the diesel-only base case, which emits approximately 3,983,094 kg CO₂/year, the hybrid system reduces emissions to 1,948,927 kg CO₂/year, achieving a total annual GHG reduction of 2,034,167 kg CO₂. This represents a 51% reduction in CO₂ emissions, in addition to reductions in carbon monoxide, nitrogen oxides, hydrocarbons, and sulphur oxides.

Fuel combustion contributes to system emissions, with annual CO₂ emissions totalling 1,948,927 kg. Additional emissions include carbon monoxide (1320 kg), hydrocarbons (44.2 kg), sulphur oxides (4837 kg), and nitrogen oxides (12,433 kg). These findings emphasise the environmental benefits of integrating renewable energy sources into hybrid systems to mitigate carbon emissions.

Table 4 Results of the sensitivity analysis

ABB-ME2 Capital Cost Multiplier (*)	C56U-330P Capital Cost Multiplier (*)	Diesel Fuel Price (\$/L)	C56U-330P (kW)	CAT-1000 (kW)	ABB-ME2	ABB-PSC (kW)	Dispatch	Cost/NPC (\$)	Cost/COE (\$)
0.25	0.25	1.4	2000	800	43,758	1065,348	CC	5.71 M	0.240779
0.5	0.25	1.4	2000	800	43,560	1077,361	CC	5.89 M	0.248172
0.75	0.25	1.4	2000	800	43,758	1065,685	CC	6.05 M	0.255139
0	0.25	1.4	2000	800	43,758	1065,348	CC	5.54 M	0.2336
1	0.25	1.4	2000	800	43,758	1065,685	CC	6.22 M	0.262319
0.25	0.5	1.4	2000	800	43,758	1065,348	CC	5.75 M	0.242255
0.5	0.5	1.4	2000	800	43,560	1077,361	CC	5.92 M	0.249648
0.75	0.5	1.4	2000	800	43,758	1065,685	CC	6.09 M	0.256615
0	0.5	1.4	2000	800	43,758	1065,348	CC	5.58 M	0.235076
1	0.5	1.4	2000	800	43,758	1065,685	CC	6.26 M	0.263794
0.25	1	1.4	2000	800	43,758	1065,348	CC	5.82 M	0.245206
0.5	1	1.4	2000	800	43,560	1077,361	CC	5.99 M	0.252599
0.75	0.5	1	2000	800	43,758	1065,685	CC	4.86 M	0.205098
0	0.5	1	2000	800	43,758	1065,348	CC	4.35 M	0.183559
1	0.5	1	2000	800	43,758	1065,685	CC	5.04 M	0.212278
0.25	0.75	1	2000	800	43,758	1065,884	CC	4.56 M	0.192216
0.5	0.75	1	2000	800	43,758	1065,685	CC	4.73 M	0.199395
0.5	1	1	2000	800	43,758	1065,685	CC	4.76 M	0.20087
0.75	1	1	2000	800	43,758	1065,685	CC	4.93 M	0.20805
0	1	1	2000	800	43,758	1065,348	CC	4.42 M	0.18651
1	1	1	2000	800	43,758	1065,685	CC	5.11 M	0.215229

3.6 Sensitivity analysis

HOMER Pro was used to conduct sensitivity analyses, evaluating the impact of fuel price variations, PV capital cost multipliers, and storage capital cost multipliers on system feasibility. The optimal system was simulated under multiple sensitivity scenarios to determine the most economically viable configuration.

As shown in Table 4, fuel prices were tested at \$1.8, \$1.6, \$1.4, and \$1 per litre, reflecting fluctuations in Nigeria's diesel market. Additionally, PV and battery capital cost multipliers were analysed due to their significant influence on system economics.

A reduction in ABB-ME2 and CS6U-330P capital cost multipliers to 0.25 resulted in an NPC of \$5.71 million and a COE of \$0.241/kWh. However, as multipliers increased to 1 and diesel prices rose to \$1.4 per litre, the NPC increased to \$6.26 million, and the COE rose to \$0.264/kWh. If diesel prices were reduced to \$1 per litre, NPC and COE declined significantly to \$4.42 million and \$0.187/kWh, respectively.

Government subsidies for diesel and system components could substantially lower implementation costs. Despite the high initial capital investment, the proposed hybrid system exhibits favourable economic indicators, including a payback period of 1.53 years, an IRR of 65%, and an ROI of 60%.

The sensitivity analysis evaluated diesel prices at \$1.0, \$1.4, \$1.6, and \$1.8 per litre, and PV/battery capital cost multipliers ranging from 0.25 to 1.0. The analysis revealed that reducing diesel prices from \$1.6 to \$1.0/L decreased NPC from \$5.71M to \$4.42M and COE from \$0.241 to \$0.187/kWh; increasing PV and battery capital cost multipliers raised NPC and COE, with the highest tested combination yielding \$6.26M NPC and \$0.264/kWh COE. The system remained economically viable across most tested scenarios, but government subsidies or falling technology prices would significantly enhance feasibility. This study confirms that HOMER is not only computationally efficient but also more transparent in handling hybrid systems when compared with alternatives like RETScreen or MATLAB-based custom models. Its simulation of dispatch strategies and lifecycle economic metrics enables decision-makers to evaluate trade-offs between renewable penetration, fuel savings, and cost. The results validate HOMER as a pragmatic and effective tool for energy planning in settings with limited data and technical capacity.

4 Conclusion

This study presents a techno-economic evaluation of a hybrid PV-diesel-battery energy system for the College of Education, Akwanga, Nasarawa State, Nigeria. Given the institution's unreliable power supply and high dependence on diesel generators, integrating renewable energy offers a viable solution for sustainable electricity generation. Using HOMER software, the optimised system configuration consists of a 32-kW solar PV array, a 1000 kVA diesel generator, batteries, and a converter, designed to meet the institution's energy demand efficiently.

The results demonstrate that the cycle charging (CC) dispatch strategy significantly improves system performance, reducing the net present cost (NPC) to \$57.1 million and the cost of energy (COE) to \$0.241/kWh, compared to the diesel-only system with an NPC of \$99.5 million and a COE of \$0.419/kWh. Additionally, the hybrid system achieves a 55% renewable fraction, leading to lower fuel consumption and GHG emissions. Sensitivity analysis reveals that diesel price fluctuations and component costs influence system feasibility, emphasising the need for strategic investment in renewable energy.

By optimising hybrid energy systems, this study contributes to enhancing energy security, reducing operational costs, and promoting environmental sustainability in educational institutions. It underscores the potential of hybrid renewable systems as a long-term solution for Nigeria's energy challenges. The originality of this work lies in its use of field-based institutional energy data, local market calibration, and scenario-based strategy comparison. These findings provide new insights for designing context-specific hybrid systems in underserved educational environments.

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Author contributions A.K.: Conceptualisation, Methodology, Validation, Formal Analysis, Writing—Review and Editing, Supervision, Project Administration. A.A.N.: Methodology, Validation, Formal Analysis, Draft Preparation, Data Curation, Writing—Review and Editing. M.A.I.: Conceptualisation, Validation, Formal Analysis, Review.

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Data availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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